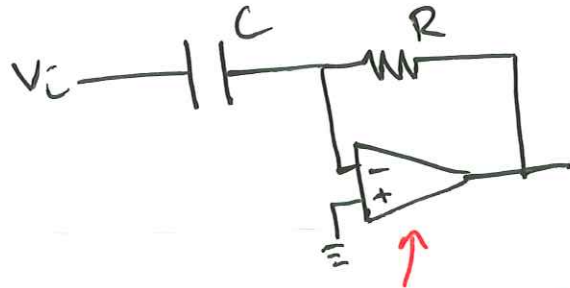


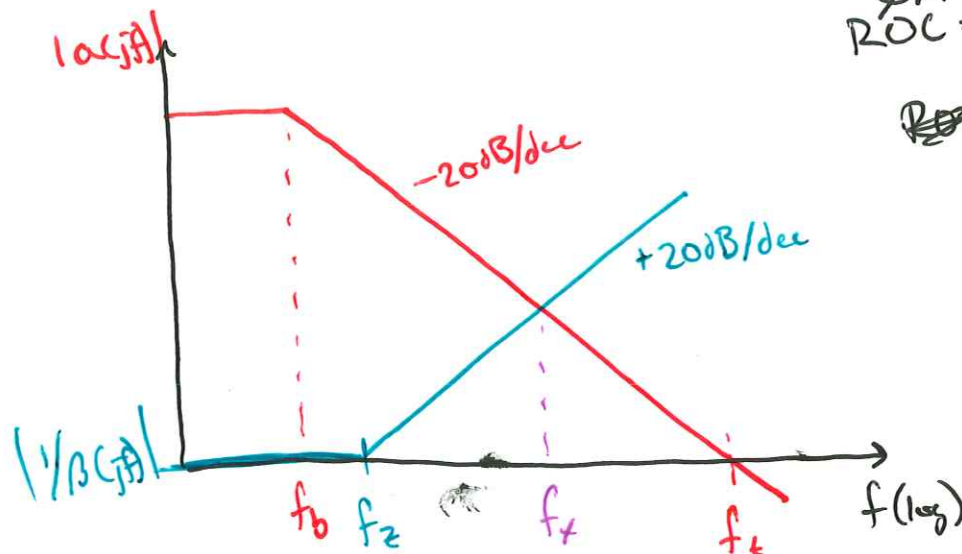
Exam 2 Next Monday! 3/30

No lecture ~~this~~ Thursday - use time to work on Checkpoint #2!
or use time as general office hours.

Differentiator



Assume GBWP Amp
(dominant-pole Amp)



(~~an~~ inverting amp)

$$\beta = R_1 / (R_1 + R_2) \rightarrow 1/\beta = 1 + R_2/R_1$$

Modify above to use C instead of R.

$$1/\beta = 1 + R/z_c = 1 + R/1/j\omega C = 1 + j\omega RC = 1 + j2\pi f RC$$

$$1/\beta \rightarrow 1 \text{ zero, no poles! } f_z = 1/2\pi RC$$

$$1/\beta = 1 + jf/f_z$$

$$A_{ideal} = -\frac{R}{z_c}$$

$$A_{ideal} = j\omega RC = jf/f_z$$

Differentiator or
~~integrator~~

$$\phi_m = 180^\circ + 45^\circ (\text{slope of } |A|) - 45^\circ (\text{slope of } 1/\beta)$$

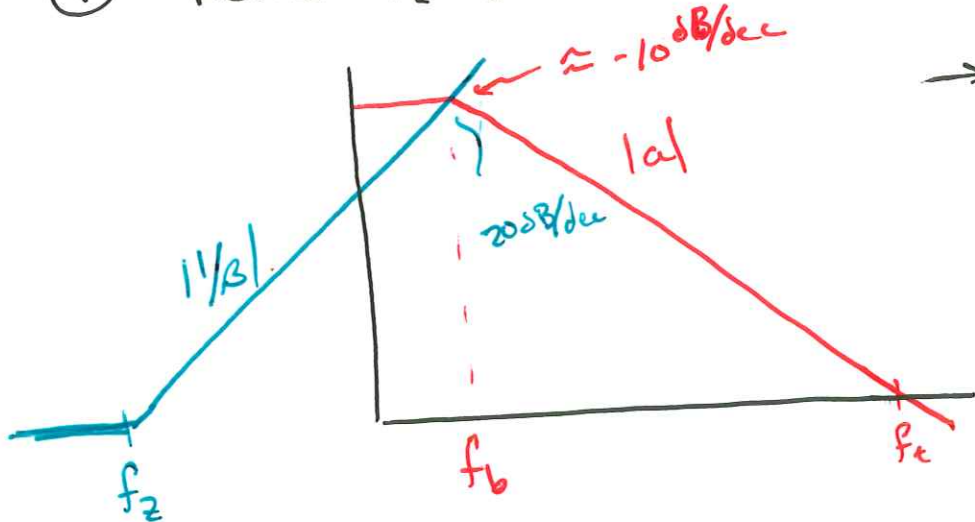
$$\phi_m = 180^\circ - 90^\circ - 90^\circ = 0^\circ$$

unstable!

$$f_x = \sqrt{f_z f_t}$$

Stabilizing Differentiator:

- ① Reduce f_z such that $f_x \leq f_b$



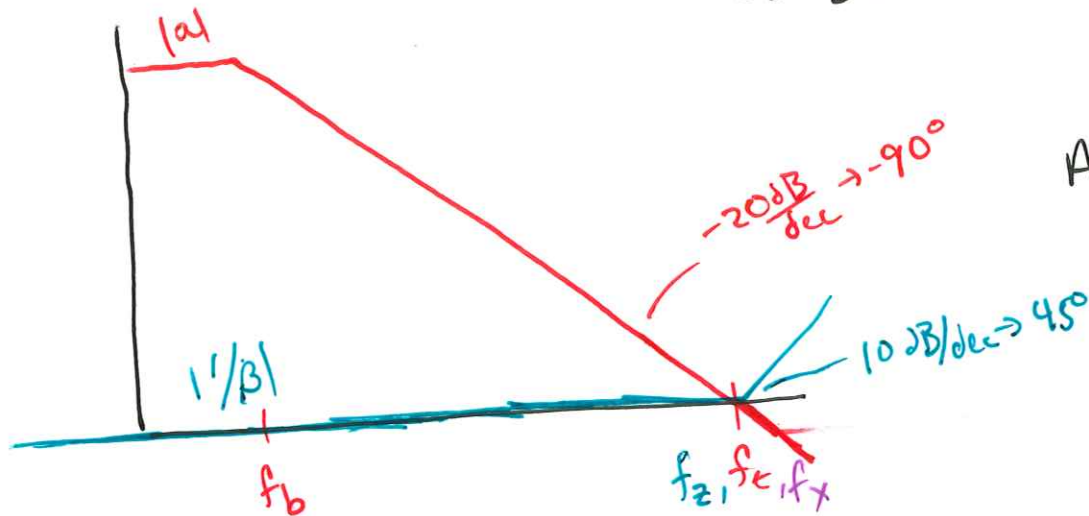
$$f_z = 1/2\pi RC$$

→ Increase C substantially!
 Since f_b is small ($\approx 10 \text{ Hz}$)
 then f_z is very small

ROC: $\phi_m = 180 - 45 - 90 = 45^\circ$
 $\uparrow 4.5 \cdot 10 \text{ dB/dec}$ $\uparrow 4.5 \cdot 20 \text{ dB/dec}$

Aideal → very large at low f ,
 & likely to saturate!

- ② Increase f_z such that $f_x \geq f_c$



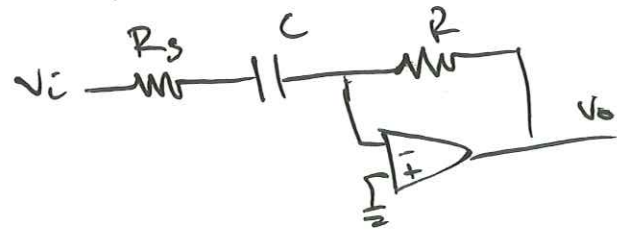
→ Decrease C substantially

$$\phi_m = 180 - 90 - 45 = 45^\circ$$

Aideal → very small at lower f
 not useful (?)

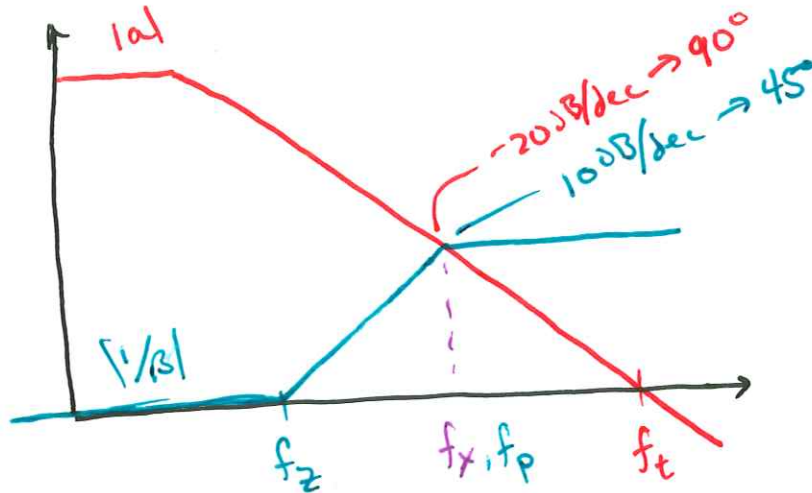
③

Add input resistor (Also required to curtail high-f gain)



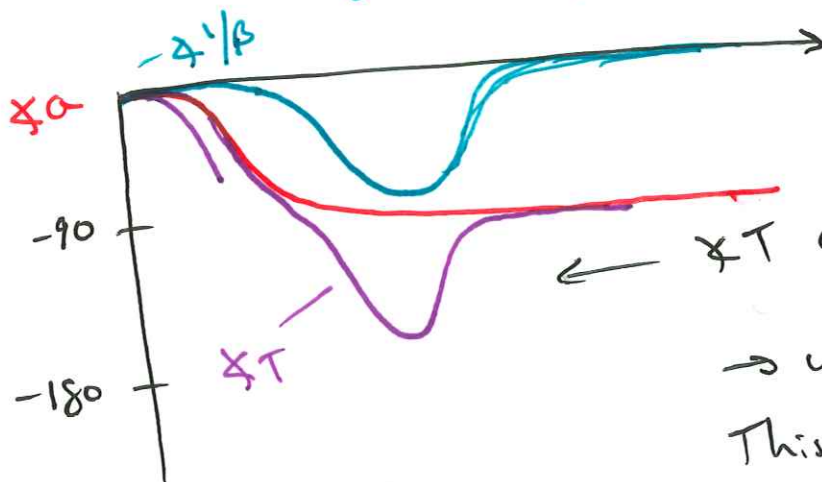
$$\begin{aligned} \frac{1}{\beta} &= 1 + \frac{R}{(R_s + 1/j\omega C)} \\ &= \frac{1 + j2\pi f(R+R_s)C}{1 + j2\pi fR_sC} \end{aligned} \rightarrow f_z = \frac{1}{2\pi(R+R_s)C} \rightarrow f_p = \frac{1}{2\pi R_s C}$$

if $R_s \ll R$
such that $f_z \approx \frac{1}{2\pi RC}$
and $f_z \ll f_p$



$$\phi_m = 180 - 90 - 45 = 45^\circ$$

$$\begin{aligned} f_p &= f_x = \sqrt{f_z f_t} \\ \downarrow \\ R_s &= R / \sqrt{f_t / f_z} - 1 \end{aligned}$$

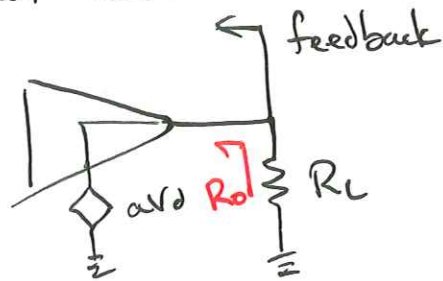


← ϕ_T approaching -180° when $|T| \gg 1$

→ will get significant gain peaking / oscillations
This does not occur in solutions ① or ②

Loading of Amplifier

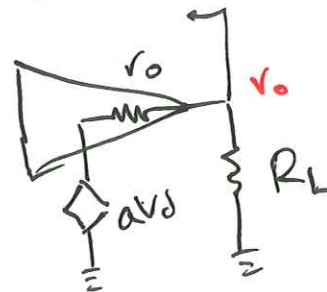
ideal case:



$$R_o \rightarrow 0$$

R_L has no effect

Non-ideal



$R_o \neq 0 \Rightarrow v_O$ affected by R_L

Example using [non]-inverting amplifier

$$\beta = \frac{R_L \parallel (R_1 + R_2)}{(R_L \parallel (R_1 + R_2)) + r_o} \cdot \frac{R_1}{R_1 + R_2}$$

if $r_o \rightarrow 0$, $\beta \rightarrow \frac{R_1}{R_1 + R_2}$

→ no phase contribution, $\therefore$ stable with GBWP amps!
 β will be lower with $r_o \neq 0$, $R_L \neq \infty$
 "slightly"

what if $R_L \rightarrow C_L$

$$\beta = \frac{Z_L \parallel (R_1 + R_2)}{Z_L \parallel (R_1 + R_2) + r_o} \cdot \frac{R_1}{R_1 + R_2}$$

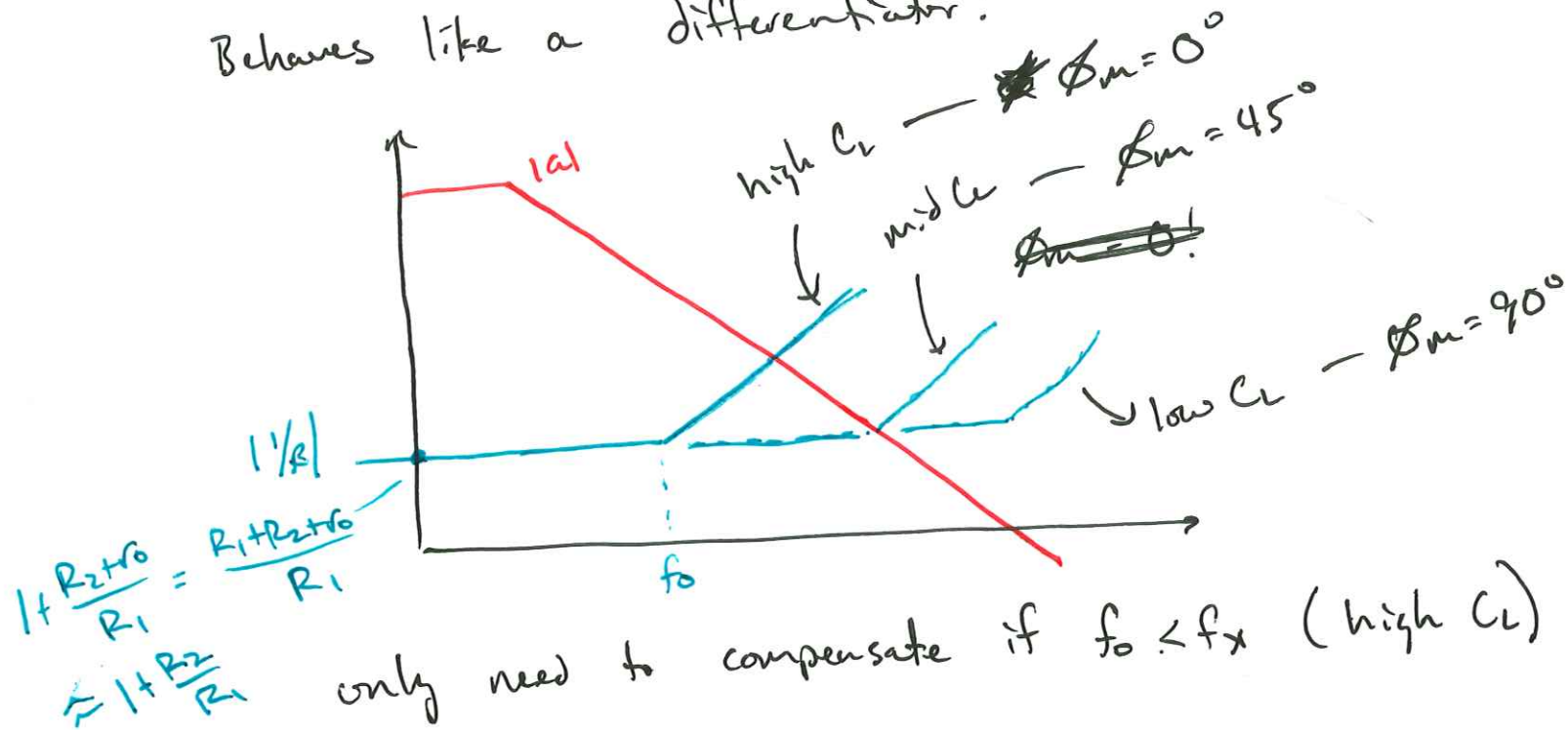
$$Z_C = 1/j\omega C_L$$

$$\beta(jf) = \frac{R_1}{R_1 + R_2 + r_o} \cdot \frac{1}{1 + jf/f_0}$$

$$f_0 = \frac{R_1 + R_2 + r_o}{2\pi f_o (R_1 + R_2) C_L}$$

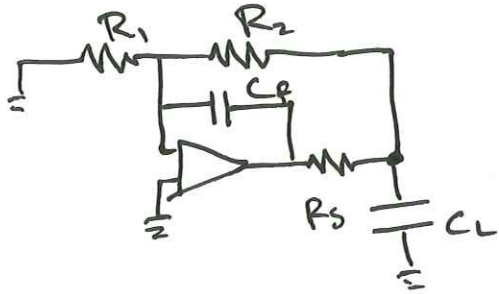
$$1/\beta = \frac{R_1 + R_2 + r_o}{R_1} (1 + jf/f_0)$$

Behaves like a differentiator!



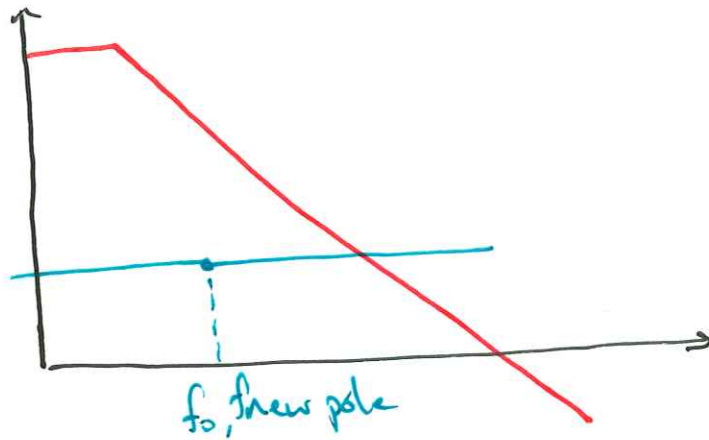
G_L loading Solutions:

① Add a feedback capacitor $\rightarrow$ Adds a pole in $1/\beta$, can cancel out zero produced by C_L

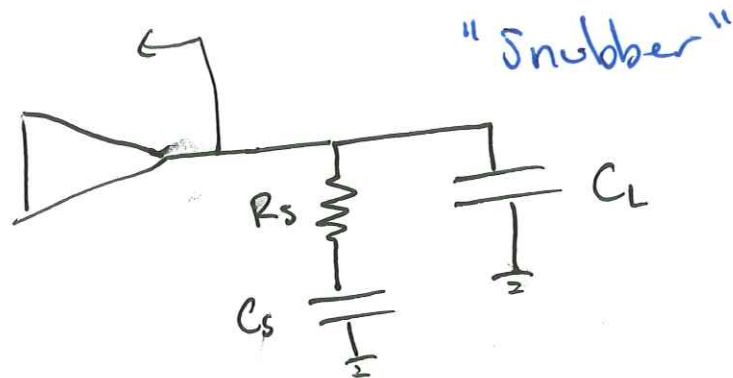


$\rightarrow R_S \& C_f$ added.

select $R_S \& C_f$ such that ~~$f_{\text{pole}} = f_0$~~ $f_{\text{pole}} = f_0$



- ② Add a parallel load to C_L to reduce capacitive effects.



want $C_S \gg C_L$ such
that load is $\approx C_L \parallel R_S @ f_x$
 $|R_S| \approx |1/j\omega C_L|$

(looks similar to differentiator solution 3)

How to implement:

- 1) start with extremely large C_S
- 2) select R_S such that operational needs are met
(ex. Gain peaking, overshoots...)
- 3) Reduce C_S until needs start deviating
→ large C_S will affect low- f operation
(R_S impactful at low- f , undesired)

Cannot do for design project!

Specifically: Breaks V-I mode.

Exam 2 topics:

DC Noise

AC Analysis (no stability)

DC Noise:

I_B , I_{OS} effects.

how to
model values
(superposition)

$I_B \rightarrow$ fixed # from manufacturers (datasheet)

"Bias current"

$I_{OS} \rightarrow$ ~~the~~ current bias error between $+$ & $-$ terminals

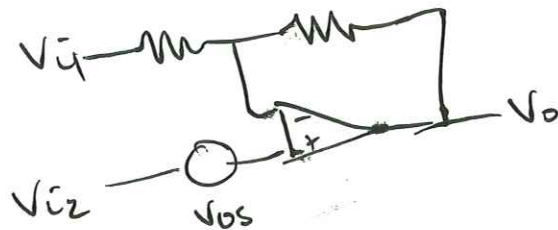
"offset current". This is a Range!

typically given as $\pm$

$V_{OS} \rightarrow$ Input offset voltage, things that contribute to it:

- Temperature changes (e.g. baseline ~~the~~ value @ temp)
- Common mode Amplification, CMRR
- Power supply variability, PSRR
- finite gain effects

V_{OS} : Always on $+$ terminal.



Error induced by V_{OS}

$$\rightarrow V_{OE}, E_O = \frac{1}{\beta} V_{OS}$$

↑ or idealized version.

AC Analysis:

$$T(jf) = a(jf)\beta(jf)$$

$$\beta(jf) = -\frac{V_O}{V_I}$$

if purely resistive, then
 ~~$\beta(jf)$~~ $\beta(jf) = \beta_0$

Dominant pole Amp:

$$a(jf) = a_0 / (1 + jf/f_b)$$

$$A(jf) = A_{ideal} / (1 + 1/T(jf)) + \underbrace{\cancel{a(jf)} \frac{a_{fe}(jf)}{1 + T(jf)}}_{\rightarrow \text{generally ignore this!}}$$

$$A(jf) = A_0 / (1 + jf/f_b) \quad \leftarrow \text{only if } \beta \text{ not dependent on } f.$$

Bunch of equations that derive from these. -- see lec. 10, 1st slide

important: $f_t = a_0 f_b$

$$f_B = (1 + a_0 \beta_0) f_b \approx f_t \beta_0$$

$$A_0 = \frac{A_{ideal}}{1 + 1/a_0 \beta_0}$$

$A_0 \approx A_{ideal}$
 if L or T
 large at $f \rightarrow 0$

$1/\beta_0$ intersection with $|a(jf)|$ determines f_B

